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Heterogeneous oxygenation of wetland soils with increasing inundation: Redox potential, water depth, and preferential flow paths

Chayan Lahiri¹  | Gregg R. Davidson²

¹Department of Biology and Geosciences,
Adams State University, Alamosa, Colorado

²Department of Geology and Geological
Engineering, University of Mississippi,
Starkville, Mississippi

Correspondence

Chayan Lahiri, Department of Biology and
Geosciences, Adams State University,
Alamosa, CO 81101.
Email: chayanlahiri@adams.edu

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Abstract

Saturated wetland soils are typically depleted in oxygen due to rapid consumption by biological processes. Fine-grained sediments limit the vertical flux of oxygenated surface water. In forested wetlands potential exists for advective transport along preferential flow pathways created by decomposing roots or tree remains, resulting in isolated zones of oxygenated pore water. This potential is enhanced in wetlands that are perched above the regional water table, where increases in surface-water depth create significant downward hydraulic gradients. Along the Mississippi River floodplain, many wetlands are now perched as a result of groundwater mining. This study was conducted at Sky Lake, MS, through a growing season, with high temporal-resolution monitoring of redox potential as a proxy for the delivery of oxygenated surface water at 30 and 60 cm depths at six locations. Results showed reducing conditions at all locations under moderate inundation, but redox potential rose significantly at a subset of locations when surface-water exceeded one meter in depth. Rising redox potential is argued to represent localised advective transport of oxygen in excess of the rate of biological consumption. Wetland plants growing in perennially saturated soils may benefit from periodic increases in water depth and subsequent increase in the flux of oxygen along preferential flow paths.

KEYWORDS

redox potential, wetland soil, inundation, advection, surface-groundwater interaction, oxbow lake, bald cypress, Mississippi River floodplain

1 | INTRODUCTION

Under typical saturated conditions, soil-zone respiration and degradation of organic matter rapidly consume available oxygen, leaving the subsurface anaerobic. Depletion of oxygen can take place within a few millimetres below the sediment surface (Gardiner & James, 2012; Howeler, 1972; Howeler & Bouldin, 1971). Studies of soil-zone oxidation–reduction (redox) potential almost invariably report reducing conditions beginning soon after inundation and continuing until soils once again become aerated (e.g. D'Amore, Stewart, & Huddleston, 2004; Dorau & Mansfeldt, 2016; Eshel & Banin, 2002; Fiedler, 2000;

Mansfeldt, 2003; Moorberg, Vespraskas, & Niewoehner, 2018; Unger, Rose-Marie, Motavalli, & Kabrick, 2008). Exceptions have been reported only where water is advectively flowing through the soil zone, delivering oxygenated water faster than it is consumed by biological decay and respiration. Examples include a reservoir study in which oxygenated groundwater upwelling through a lake bottom was first recognized by redox potentials in the lake sediments that unexpectedly increased with depth (Davidson, Bennett, Beard, & Waldo, 2005), and a study of water fluxes in the hyporheic zone during a hurricane, where heavy rains resulted in localised zones of rapid groundwater flux (Sawyer, Kaplan, Lazareva, & Michael, 2014).

In forested wetlands, there is substantial heterogeneity created by roots and fallen trees, providing potential pathways of higher hydraulic conductivity to connect surface water and underlying groundwater. Transport of surface water along preferential pathways could produce pockets of oxygenated pore water within otherwise anoxic soils. Oxygenation under saturated conditions by this mechanism was suggested by Stahle and Cleaveland (1992) as a possible explanation for why some bald-cypress in perennially-saturated soils experience increased growth when it rains (rising water increasing the vertical hydraulic gradient and delivery of oxygenated surface water). Though wetland plants are adapted to low-oxygen environments, many nonetheless experience more robust growth under more oxidizing conditions. Oxygen is known to facilitate nutrient uptake by plant roots (Dickson & Broyer, 1972; Dickson, Broyer, & Johnson, 1972; Pezeshki & Santos, 1998), and some wetland species pump oxygen into the rhizosphere (Chabbi, McKee, & Mendelssohn, 2000; Laskov, Horn, & Hupfer, 2006; Mendelssohn, 1993). Downward flux of surface water may also be enhanced as water is withdrawn by root uptake and transpiration (Jaffe, Kallin, & Smith, 1998).

An earlier study in the forested wetland of Sky Lake, MS (the location of the present study) employed water chemistry to document the presence of preferential flow pathways in the top 3 m of sediment (Davidson, Laine, Galicki, & Threlkeld, 2006). Rising surface-water levels resulted in rapid changes in oxygen-isotope signatures and chloride concentrations in pore waters at isolated depths, consistent with localised downward movement of water in response to an increase in the vertical hydraulic gradient. More recently, Gratzner et al. (2020) documented formation of a temporary mound in the potentiometric surface of the aquifer directly beneath Sky Lake following a 4-m rise in lake level. Response times were shorter and the magnitude of change was greater in piezometers directly beneath the wetland than in wells farther removed from the lake, providing a clear indication of direct flux of surface water to the underlying aquifer.

Oxygen concentration in sediment pore water is difficult to directly monitor over time due to electrode expense and biofouling. Redox potential, or Eh, is much easier to measure. Redox potential is measured as a voltage potential difference between the system being monitored and a standard reference electrode (Reddy & Delaune, 2008). There is no strict relationship between oxygen content and Eh, but Eh measurements can provide an indication of whether the environment is becoming more or less oxidizing, consistent with rising or falling oxygenation. Redox measurements thus have the potential to qualitatively identify fluxes of oxygenated water, or changes in oxygen delivery rates, into the subsurface (Mansfeldt, 2003).

Numerous studies have been conducted in recent years addressing specific aspects of redox measurement in soils. Studies include the importance of sealing the junction between the Cu wire and Pt tip in hand-made electrodes (Fiedler, Vepraskas, & Richardson, 2007), tests of how long electrodes may be deployed in the field without cleaning or replacement (Austin & Huddleston, 1999; Mansfeldt, 2003), possible inconsistencies between manual and automated field measurements (Rabenhorst, Hively, & James, 2007), differences between instantaneous measurement of electric potential versus

allowing equilibration time between the reference and Pt-electrodes (van Bochove, Beauchemin, & Thériault, 2002), equilibration times after removal and reinsertion (Eshel & Banin, 2002), and the number of electrodes necessary to characterize the redox potential of a soil horizon (Fiedler et al., 2007; Vepraskas & Faulkner, 2001).

The need for multiple electrodes arises from observations that measured redox potential in soils can significantly vary over millimetre scales (Fiedler et al., 2007; Yang, Hu, & Bu, 2006). Differences tend to be greater vertically than horizontally within the same layer (Fiedler et al., 2007; Yang et al., 2006), with a lesser degree of variation occurring under conditions of continuous saturation (Fiedler et al., 2007; Mansfeldt, 2003; Thomas, Miao, & Sindhoj, 2009). To account for natural variability, researchers recommend anywhere from 3 to 10 electrodes per depth to quantify the range (D'Amore et al., 2004; Dorau & Mansfeldt, 2016; Fiedler et al., 2007; Vepraskas & Faulkner, 2001). Characterising the range is important if assessing the type of redox reactions that may be occurring at a particular depth or location, such as understanding what minerals or ions may be acting as oxidizing agents. It is less critical if employing redox potential simply to document perturbations to the system, such as the introduction of oxygenated water into a reducing environment (e.g. Sawyer et al., 2014). Clustered electrodes commonly vary within a range of about 100 mV, whereas an oxygenation event will typically raise the Eh by several hundred mV. Individual electrodes in a cluster generally follow the same overall pattern of change (Fiedler et al., 2007). In a soil-redox study in a natural wetland, D'Amore et al. (2004) noted that variability among the electrodes at each depth did not change the interpretation of the redox environment, so only average Eh values were reported.

In the current study, high temporal-resolution measurements were employed in a forested wetland at multiple locations over a full growing season to test two related hypotheses. Hypothesis 1: Vertical preferential flow pathways transect the fine-grained channel fill deposits connecting wetland surface water to the underlying sand aquifer. Hypothesis 2: Advective delivery of oxygen, in response to increases in surface water depth, exceeds respiration, and decomposition demands in isolated pockets of the subsurface even under prolonged periods of inundation. Evidence in support of the two hypotheses was expected to include (a) high redox-potential variability at the same depth at different wetland locations in response to changing surface water levels, and (b) one or more electrodes recording increases or anomalously high redox potential in spite of prolonged inundation.

2 | STUDY AREA

Sky Lake is an oxbow lake-wetland system located in the floodplain of the Mississippi River, ~10 km north of Belzoni, MS (Figure 1). The oxbow was once part of the Mississippi-Ohio River system (Thorne & Curry, 1983) and is between 3800 and 5000 years old (Wren, Davidson, Walker, & Galicki, 2008). The lake is surrounded by a forested wetland up to 1 km in width. The outer edge of the oxbow wetland sits ~3 m below the floodplain and drops gradually by an additional

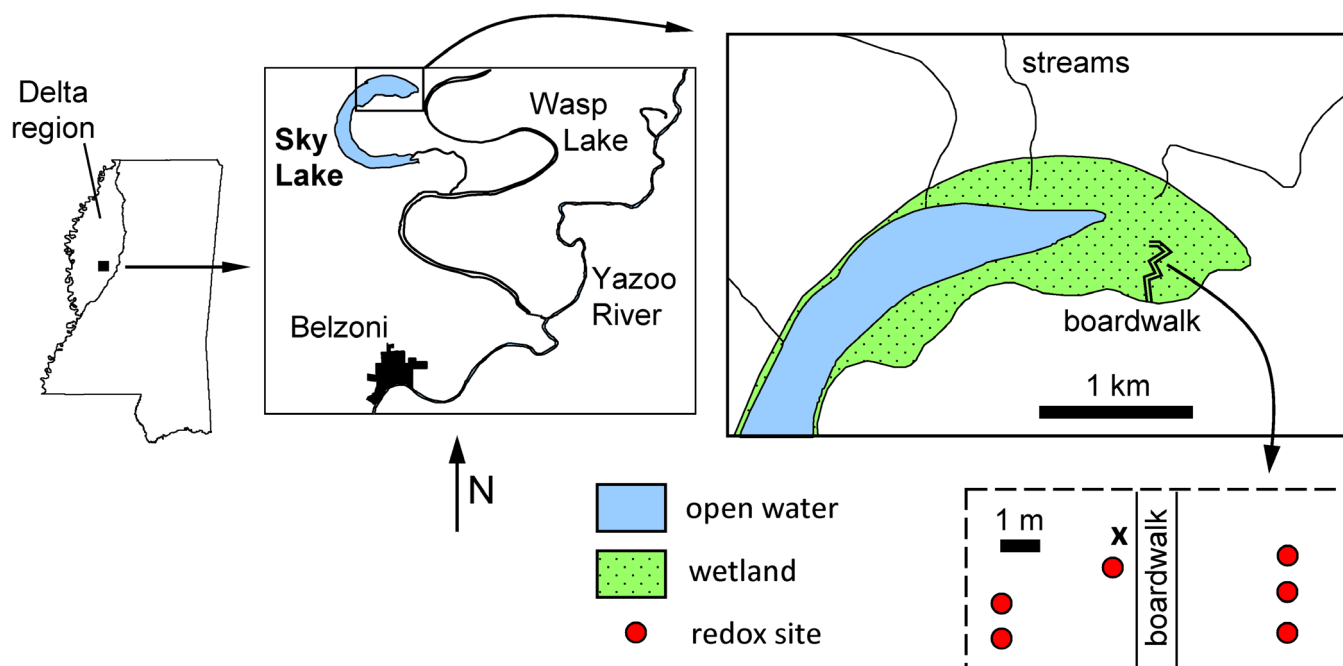


FIGURE 1 Sky Lake with enlargement of the northern wetland, and layout of the experimental area. Two Eh electrodes were placed at each location (30 and 60 cm depth) marked with circles. The data logger was mounted to the boardwalk at the X. The Delta region is the portion of the Mississippi River floodplain lying inside the state boundary. The southern and eastern extent of the lake are located at latitude $33^{\circ}15'N$ and $90^{\circ}30'W$, respectively

meter towards the inside edge at the open water line. The lake bottom is nearly flat. Sky Lake receives runoff from the surrounding agricultural land and backflow from Wasp Lake and the Yazoo River during high-water events (Galicki, Davidson, & Threlkeld, 2008a). The wetland is dominated by bald cypress (*Taxodium distichum*), including some of the oldest and largest bald cypress trees in the state of Mississippi (Galicki et al., 2008b). Wetland sediments are predominantly silty clay and rich in organic matter (Galicki et al., 2008a). The regional groundwater table was probably in direct connection with the wetland historically, but long-term pumping for irrigation has drawn the water table below the bottom of the lake and wetland. Water levels reported by Gratzner et al. (2020) for a series of wells and piezometers in and around Sky Lake document the lake and wetland are perched above the underlying aquifer. The study site for redox measurements was located adjacent to an elevated boardwalk at the northern end of the oxbow (Figure 1). (Piezometer SLW-1, reported by Gratzner et al. (2020) was installed at this site at a later date.)

3 | METHODS

Redox potential data were collected March through December 2013. An elevated boardwalk provided easy accessibility to the centre of the wetland and a structure for mounting sensitive electronic equipment above the height of typical maximum inundation. Water levels often exceed 3 m at the site. The redox electrodes were constructed in accordance with the steps outlined by Reddy and DeLaune (2008), welding 1 cm of platinum wire to 12-gauge copper wire. The

connection between the copper wire and platinum tip was sealed in epoxy and heat-shrink plastic tubing. Electrode cables were 10 m in length and connected to a data logger at the boardwalk. A commercial Ag:AgCl reference electrode was used with a 3.8 M KCl salt bridge, with measured voltage potentials corrected relative to the standard hydrogen electrode (reported as Eh). Electrodes were tested using a standard Zobel solution at the beginning and end of deployment.

Six electrodes were installed at each depth (30 and 60 cm). Multiple electrodes were not clustered together at each measurement location for several reasons. (a) The primary objective was documenting relative changes that accompany introduction of oxygenated water into reducing soils. Absolute values necessary for characterizing specific redox reactions were not essential. (b) As described in the Introduction, changes accompanying oxygenation events generally exceed typical small-scale spatial variation in redox potential, especially under conditions of continuous saturation. (c) Preferential flow pathways are discrete features that can be narrow and sinuous. Multiple electrodes even a few centimetres apart are unlikely to all be installed in the same flow feature.

Care was taken to select locations, with one exception, beyond the area of ground disturbance created by construction of the boardwalk. Trees and woody growth precluded placement in a regular rectangular grid. Instead, locations were selected across two transects on either side of the boardwalk, with roughly 1 m spacing. At the beginning of the study, small-diameter starter holes were made with a metal rod at a 45 degree angle (orthogonal to the boardwalk), with surface-entry positions adjusted so the 60 cm target depth aligned vertically beneath the 30 cm depth. For each electrode, a 10-m long,

2-cm diameter PVC tube was run from the boardwalk to the wetland location and pushed to the target depth with a rubber stopper in the end. Insertion by direct push ensured there was no gap between the sediment and the outside of the access tube. A smaller diameter tube was inserted to push the stopper at least 5 cm beyond the end of the PVC. Insulated electrodes were encased in 1-cm diameter plastic tubing to prevent crimping, with the platinum wire tip extending beyond the end. The electrode-tube was made slightly longer than the PVC tube, ensuring that the platinum wire would extend a short distance into the sediment at each target depth. This design allowed access to the electrodes for cleaning or checking during high water, eliminated significant disturbance of the sediment with removal and redeployment, and ensured that the electrodes were returned to the same spot beneath the surface each time. Mud coating the electrode tips when retracted after the first week of deployment confirmed that sediment had refilled the void created by pushing out the rubber stopper.

A Ag:AgCl reference electrode was installed in the sediments at an intermediate point relative to the measuring electrodes. The redox electrodes were removed and cleaned once every 3 weeks to prevent bio-fouling (Devitt, Stolzy, Miller, Campana, & Sternberg, 1989; Ver-shinin & Rozanov, 1983), though some authors are of the opinion that electrodes can be left in the sediment to collect data for a year or more without cleaning (Cogger, Kennedy, & Carlson, 1992; Faulkner & Patrick, 1992; Fiedler et al., 2007; Vepraskas & Faulkner, 2001). In contrast to results reported by Eshel and Banin (2002), no significant changes in Eh were observed immediately after cleaning. The absence of an elevated spike in Eh may be attributed to the unique experimental design that eliminated significant introduction of atmospheric air or soil disruption with removal and reinsertion.

Redox potential was recorded every 0.5 hr, except for a 3-week interval beginning in mid-July when the data logger malfunctioned. Water depth was measured in the oxbow lake at 4-hr intervals using nonvented data-logging pressure transducers corrected for barometric pressure by a dedicated transducer.

4 | RESULTS AND DISCUSSION

Redox potential measurements were collected for 6 months, beginning at a time when the site had already been inundated for 3 months (Figure 2). Water level reached a maximum depth of 3.7 m in late February and was in the process of a steady decline when Eh measurements began in mid-March. The site remained inundated for an additional 4.5 months, with fluctuations in water depth reaching a maximum of 2.3 m before dropping below the ground surface for the last 1.5 months of the study.

The competing processes of oxygen delivery and consumption are readily seen in the diurnal changes in redox potential. Daily changes in Eh on the order of 80 mV were observed between daytime lows corresponding to peak sunlight and plant-respiration, and nighttime highs, consistent with observations of others (e.g. Clay, Clapp, Molina, & Linden, 1990; Fiedler et al., 2007). The diurnal fluctuations also provide a reasonable indication that the electrodes were

accurately recording relative changes in redox potential over time at each location (Figure 3). Diurnal fluctuations did not compromise the ability to detect advective oxygenation events, as these result in much larger increases in Eh.

Mean daily Eh values were all initially greater than 100 mV, with some exceeding 400 mV (Figure 4), in spite of being inundated for the previous 3 months. One possible cause of high initial values is the introduction of oxidizing water at the time of electrode deployment, though this is considered an unlikely explanation. The high redox potential persisted for longer than installation-induced readings

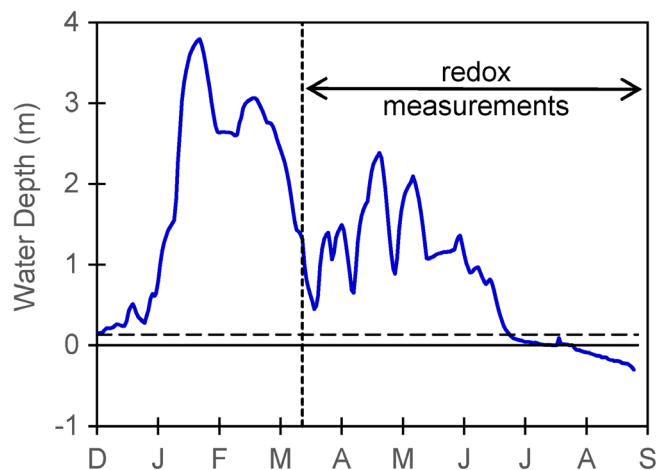


FIGURE 2 Water depth at wetland site prior to and through the 2013 growing season. Zero water depth represents the average elevation of the wetland floor. The horizontal dashed line at 13 cm represents the approximate depth at which hummocks are completely submerged

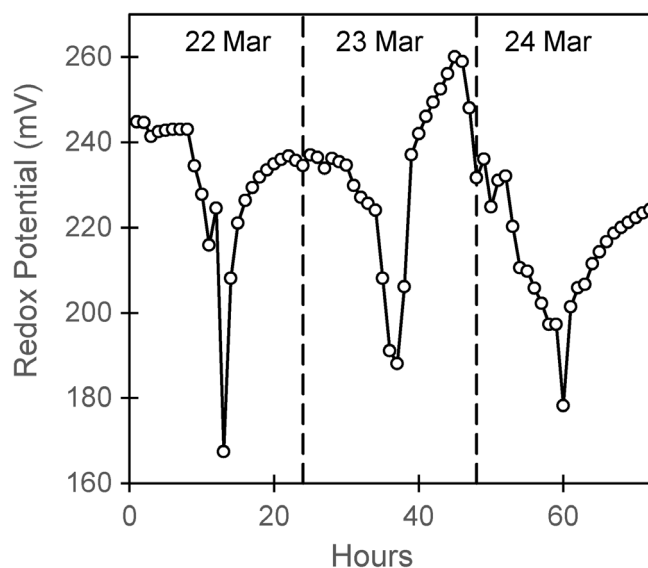


FIGURE 3 Hourly redox potential for electrode E8 (30 cm) plotted for 22–24 March. Vertical dashed lines represent midnight of each day

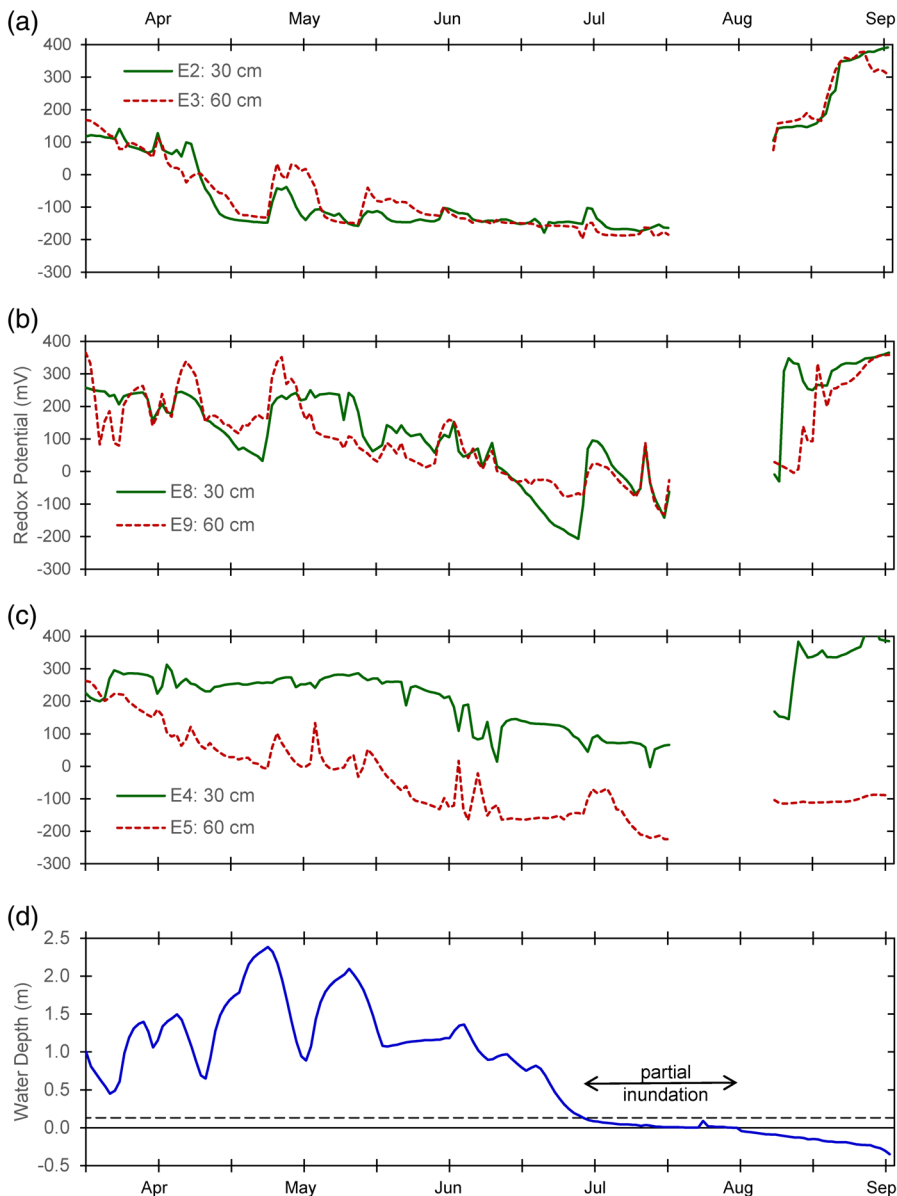


FIGURE 4 Representative time series of redox-potential responses to changing water levels. Graphs a–c correspond to groups a–c in Figure 5 (refer text for discussion). Missing data represents a period of datalogger malfunction

reported by others. Eshel and Banin (2002) and Thomas et al. (2009) reported equilibration times that required one to 2 weeks. High Eh at Sky Lake persisted at some locations for up to 2 months under continuously inundated conditions. Persistent high Eh is not likely to be caused by the installation of the electrodes, as there was no gap between the sediment and access tube, the electrode occupied most of the volume inside the tube, and the tip of the electrode extended several additional centimetres into the sediment. High initial Eh values are more likely the result of varying degrees of advective delivery of oxidizing surface water into the wetland soil prior to and during measurement. Water levels were still in decline from a peak depth of 3.7 m when Eh measurements began. Alternatively, the higher initial Eh values may simply reflect the lower biological oxygen demand prior to the start of the growing season.

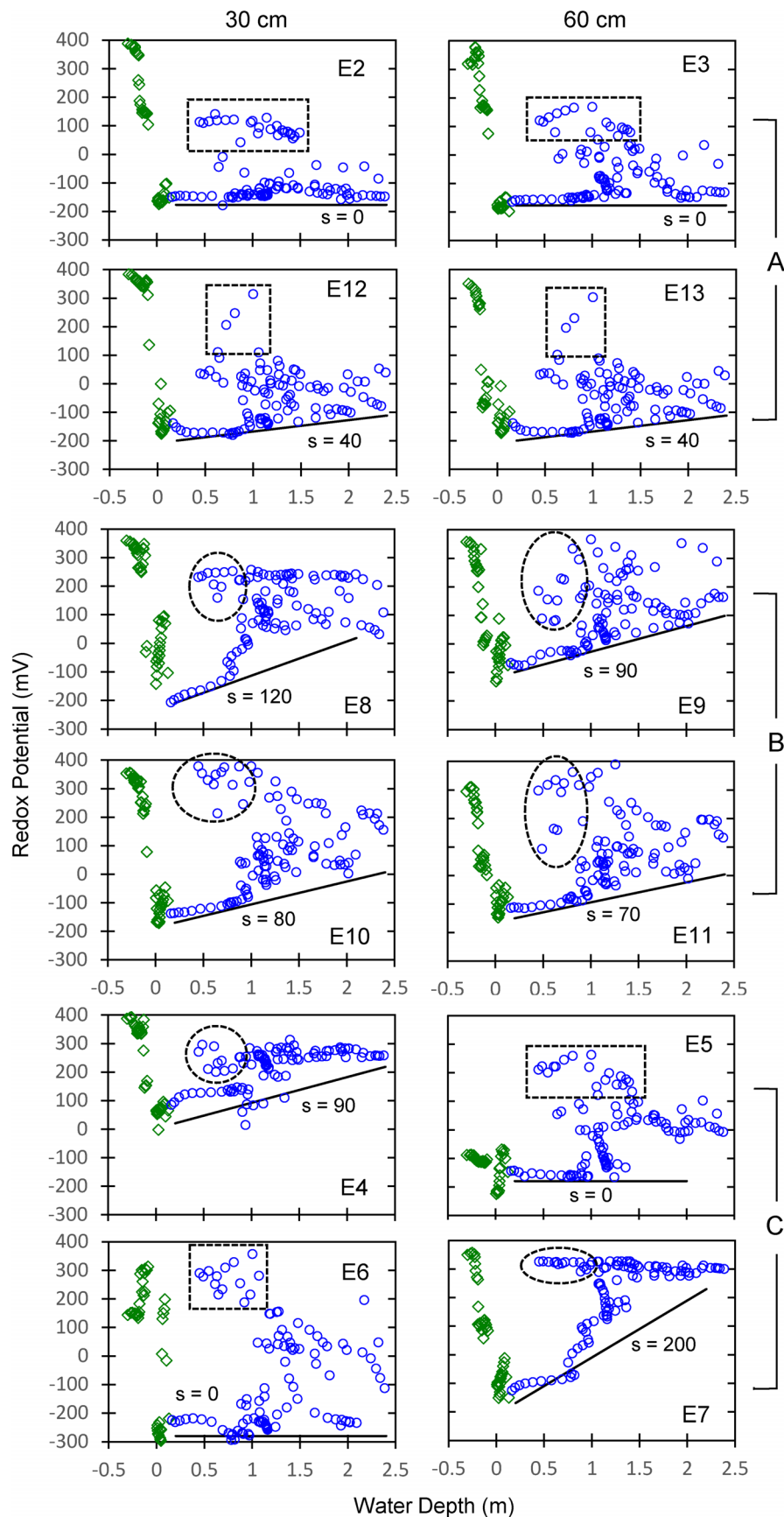
Two types of redox-potential responses were observed with respect to time and water depth. Representative time series for paired

electrodes (30 and 60 cm) are shown in Figure 4. Figure 5 shows redox potential as a function of water depth for all electrodes, with the same groupings (a–c) as Figure 4. Two different markers are used in Figure 5, one for full inundation and one for partial and full aeration (partial aeration represents transition when hummocks are above water and low spots hold ponded water). The following discussion considers Eh responses under full inundation first, then under partial and full aeration.

4.1 | Redox potential under full inundation

Characteristics of the first group (E12/E13; E2/E3 shown in Figure 4a) include low initial mean-daily redox potential, a faster overall decline to negative Eh values, minimum values in the –150 to –300 mV range, and similar values recorded between the 30 and

FIGURE 5 Water depth versus Eh. Group A has little or no correlation between redox potential and water depth under inundation. Squares identify data from the first few weeks of measurement. Group B exhibits higher redox potential with greater water depth. Circles identify data when surface water level dipped below 1 m for a brief period before rising again. Group C probes represent decoupled electrodes, where one depth in a pair behaves more like group A and the other behaves more like group B. Lines and slopes (s) indicate the trend of the lower Eh values with water depth. Green diamonds=shallow sediment aerated; Blue circles=sediments saturated [Correction added on 28 January 2020 after first online publication: the caption of Figure 5 has been updated in this version.]



60 cm depths. When plotting Eh versus water depth (Figure 5a), no apparent correlation is observed under full inundation (R^2 values < 0.02). The highest Eh values are all associated with the initial measurements following the winter high water-depths (first 2 weeks of measurement, identified with dashed box). These results are consistent with minimal vertical flow during the period of measurement, such that the rate of oxygen consumption generally exceeded the rate of advective (or diffusive) delivery.

Characteristics of the second group (E10/E11; E8/E9 shown in Figure 4b) include higher initial Eh, average Eh values remaining high for several months before beginning to decline, and minimum values mostly above -100 mV. As with the first group, similar values were recorded internally between each pair of electrodes. The data in Figure 5b show general patterns of higher Eh with water depths over 1 m, though with considerable scatter (R^2 of 0.05 to 0.18). These results are consistent with greater downward flow as water level rises, where oxygen delivery rates begin to exceed consumption rates. High Eh values at low water depth all occur when the wetland water-level dipped briefly below 1 m before rising again (identified with dashed circles in Figure 5). Time may have been insufficient for redox potentials to decline in response to short-term reductions in advective flow.

Ideally, the slope of the water depth versus Eh data (under full inundation) would provide the strongest indication of vertical flux of water and oxygen, with higher slopes associated with greater oxygenation. The approximate relationship between oxygen content and redox potential, coupled with the compounding factors mentioned above (initial high Eh; lag in redox response with brief change in water depth) can skew the best-fit line. The impact of these compounding factors, however, is minimised in the lower Eh range. In Figure 5, lines are drawn following the trend in the lower Eh values (most Eh values lying above the line). The group b electrodes exhibit a much higher slope (70–120) relative to the group a electrodes (0–40).

The third group (E6/E7; E4/E5 shown in Figure 4c) differs from the others only in that the 30 and 60 cm depths appear to be decoupled. In the E4/E5 pair, E4 (30 cm) behaves more like group b, and E5 (60 cm) more like group A. The reverse is observed in the E6/E7 pair (time series not shown), where E6 (30 cm) behaved more like group a, and E7 (60 cm) more like group b. Such behaviours are indicative of preferential flow pathways with variable orientation. Diagonal flow pathways can result from decomposing roots or tree limbs buried at odd angles (visually observed at Sky Lake). In Figure 5c, the Eh recorded by E6 and E5 (at different surface locations) are poorly correlated with water depth. High Eh is only associated with the initial readings, consistent with group a responses. The Eh recorded by E7 and E4 have higher correlations with water depth ($R^2 = 0.33$ for both), and the circles again identify high Eh with low water depth only during short-term drops in water level, consistent with group B responses.

Short-term spatial and temporal variability in soil redox potential in response to changing flow conditions has been reported by others, mostly associated with recharge events through the unsaturated zone. Sawyer et al. (2014) investigated soil redox potential in the hyporheic zone of a stream and floodplain during a hurricane event. Eh

fluctuated on the order of 300 mV at isolated locations, attributed to higher localised rates of groundwater flux along preferential flow pathways. At Sky Lake, short-term fluctuations of 200 mV were common, with a total range (under full inundation) exceeding 500 mV at some locations over the course of the growing season.

4.2 | Redox potential under partial and full aeration

Under partial and full surface aeration, there is a strong correlation (R^2 up to 0.93) between redox potential and depth to water below the surface for all but one electrode (E5, Figure 5c). The explanation for this behaviour is possibly more complex than it first appears. It is not necessarily a matter of exposing the pores around the electrodes to the air, because according to the water-level record, the 30 cm electrode was only above water the last two days of the study, and the 60 cm electrode was never aerated.

Water level was tracked in the lake adjacent to the wetland, allowing water levels to be recorded that fell below the wetland floor. As the lake level drops below the wetland, however, the wetland sites become increasingly decoupled from the surface water. Vertical drainage from the wetland soils is likely to be faster (more buried trees and limbs) than through the un-vegetated lake bottom. The actual depth of aeration below ground surface may thus be greater than reflected by the negative lake levels in Figures 4 and 5 (negative lake level meaning inferred depth below the wetland ground surface to where wetland pores remain saturated). Shallow piezometers installed in the wetland at depths of 0.6 and 1.2 m (at a different time than the Eh measurements) recorded subsurface water levels in the wetland up to 20 cm lower than lake levels during a period of wetland aeration. A 20 cm differential, however, still leaves the 60 cm electrode in saturated sediments during a period of rapidly rising Eh values.

An alternative explanation is related to the ability of wetland plants to pump oxygen into the subsurface through their roots (Mendelssohn, 1993). Once root hairs near the surface are exposed to the air, even under only partially aerated conditions, oxygen delivery to deeper roots in saturated sediments may also increase, resulting in a rise in redox potential. Further study is needed to test this hypothesis.

4.3 | Influence on wetland tree growth

Results from previous work on the relationship between bald cypress growth and surface-water depth have been equivocal, with some reporting a negative relationship (Dicke & Toliver, 1990; Middleton & McKee, 2005) and some positive (Stahle, VanArsdale, & Cleaveland, 1992; Young, Keeland, & Sharitz, 1995). It has been suggested that where wetlands lie above the local water table, increases in surface-water level may enhance downward delivery of oxygen to the root zone and facilitate growth (Davidson et al., 2006; Stahle & Cleaveland, 1992). A study of bald cypress tree rings along an elevation transect in Sky Lake (Weber, 2010) found a positive relationship between

radial growth and water depth for trees growing in permanently saturated soil, but not for trees growing at higher elevations that experienced frequent aeration (Weber, 2010). With periodic aeration, tree roots have ample access to oxygen, with variations in oxygen delivery during intervals of high and low inundation having minimal additional impact. Under permanent inundation, however, localised increases in downward advective flow at greater water depths may be the main source of oxygen. Permanently inundated trees would thus respond positively to rising water levels that induce increased flow of oxygenated surface water. Additional investigation is continuing at Sky Lake to elucidate relationships between water depth and tree productivity.

5 | CONCLUSIONS

The results of this study, coupled with the work of Gratz et al. (2020), provide strong support for the first hypothesis that preferential pathways are common through the fine-grained wetland sediments of Sky Lake. High spatially variable Eh responses to changes in water depth are more consistent with localised zones of higher or lower vertical conductivity than would be expected under uniformly static saturated conditions. The redox results also provide support for the second hypothesis, that advective downward flow in response to increases in surface-water level delivers oxygen to isolated pockets in the subsurface at a rate that exceeds the biological oxygen demand. Scatter in the data can be attributed to (a) a lack of a one-to-one relationship between redox potential and oxygen concentration, and (b) delayed or muted changes in redox potential during periods of rapidly changing water level. For perched-wetland systems (situated above the local groundwater), plants growing in perennially saturated soils may benefit from periodic increases in water level that induce greater vertical fluxes of oxygenated water.

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DATA AVAILABILITY STATEMENT

The data that supports the findings of this study are available from corresponding author upon reasonable request.

ORCID

Chayan Lahiri  <https://orcid.org/0000-0002-7454-4196>

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